

DUET: Endogenous Failure Detection and Recovery via Dual-Action Manifold Alignment

Anonymous Authors

Abstract—Vision-language-action (VLA) policies fine-tuned on limited demonstrations can struggle to detect and recover from out-of-distribution (OOD) execution deviations. Existing solutions often require external detectors or costly recovery data. We introduce DUET (Dual-manifold Unified Execution and Transfer), a dual-action-head module trained on successful demonstrations alone. Relative actions provide nominal control, while a state-free absolute head predicts RVQ-anchored waypoints for recovery. Cross-space disagreement, complemented by state-distribution distance, gates the transition between them, coupling failure detection with correction. On four LIBERO suites under five failure families, DUET improves nominal success from 93.8% to 96.9% and action-execution failure success from 27.9% to 64.6% over its relative-only baseline. Its disagreement score achieves the highest AUROC (0.845) among the evaluated success-only detectors. Transferred to a π_0 backbone on a Franka arm, the same module matches nominal success (90.0%) and raises robot-state OOD success from 10.0% to 77.5%.

I. INTRODUCTION

Vision-language-action (VLA) policies adapt pretrained vision-language models to map visual observations and language instructions to robot actions [1], [2]. Trained on successful demonstrations, they can drift during closed-loop execution until task completion becomes unreliable—an *execution failure*. Before failure becomes irreversible, the rollout may leave the observations, configurations, or action consequences supported by demonstrations. We call this an *execution deviation*. An unexpected arm displacement changes the starting pose, while a dropped object changes the robot-object configuration even if the instruction remains unchanged. Deviation and failure are not equivalent: unseen states may still be handled successfully, while compounding errors can cause failure under familiar observations. We focus on deviations that elevate failure risk but remain recoverable by returning toward demonstrated behavior.

To anticipate failures and recover from these execution deviations, existing methods often train an external failure predictor [3] and a separate recovery controller using human takeovers or expert interventions [4], [5], [6]. Their supervision remains loosely coupled: the detector raises an alarm, while the controller depends on costly corrective data. We ask whether a VLA can use its own action predictions to detect a recoverable deviation and generate a retry direction from successful demonstrations alone, without an external failure model or recovery data.

Achieving this poses two coupled challenges. Successful demonstrations teach a policy how to act along successful

trajectories, but do not explicitly teach it when its actions become unreliable or how to return after leaving those trajectories. Detection must therefore derive a meaningful risk signal from the policy itself, while recovery must infer a corrective direction from demonstrations that do not cover the encountered deviation. Our approach is motivated by a recurring observation from prior experiments: with large-scale training data, relative actions often yield the best performance by enabling stronger generalization. However, after fine-tuning on a small demonstration set, relative-action policies struggle to recover autonomously once execution enters out-of-distribution states. Absolute actions exhibit the complementary tradeoff: they generalize less effectively in the large-data regime but more readily guide execution back toward demonstrated behavior after a deviation under limited-data fine-tuning. This observation motivates DUET, which combines a relative head for nominal execution with an absolute head for trajectory re-entry. Disagreement between the relative command and the correction toward the absolute waypoint signals rising failure risk, while the waypoint itself supplies a retry direction.

The two heads learn matched future actions from each successful demonstration. The relative head receives the live robot state, whereas the absolute head excludes explicit proprioception to retain a scene-conditioned reference. Converting its waypoint into a correction from the measured pose makes the predictions comparable: they can agree along demonstrated trajectories yet separate after displacement, indicating both a mismatch and a candidate return direction.

The module attaches to shared vision-language features and is trained on StarVLA [2] in simulation and π_0 (OpenPi) [7] on hardware. At inference, a soft gate shifts toward waypoint correction as disagreement rises and restores relative control after realignment. Irreversible failures, hardware faults, and cases requiring a new task strategy remain outside its scope.

Our contributions are threefold:

- We formulate failure-aware retry using cross-space disagreement as an endogenous failure-risk signal learned without failure labels, outperforming the evaluated detectors on TPR and AUROC (§IV-D).
- We propose DUET, unifying deviation detection and waypoint-based correction while preserving smooth nominal control (§IV-C, §IV-B).
- We evaluate DUET on StarVLA in LIBERO and on a separately trained π_0 (OpenPi) controller across four Franka tasks to test its applicability across backbones (§IV-B, §IV-F).

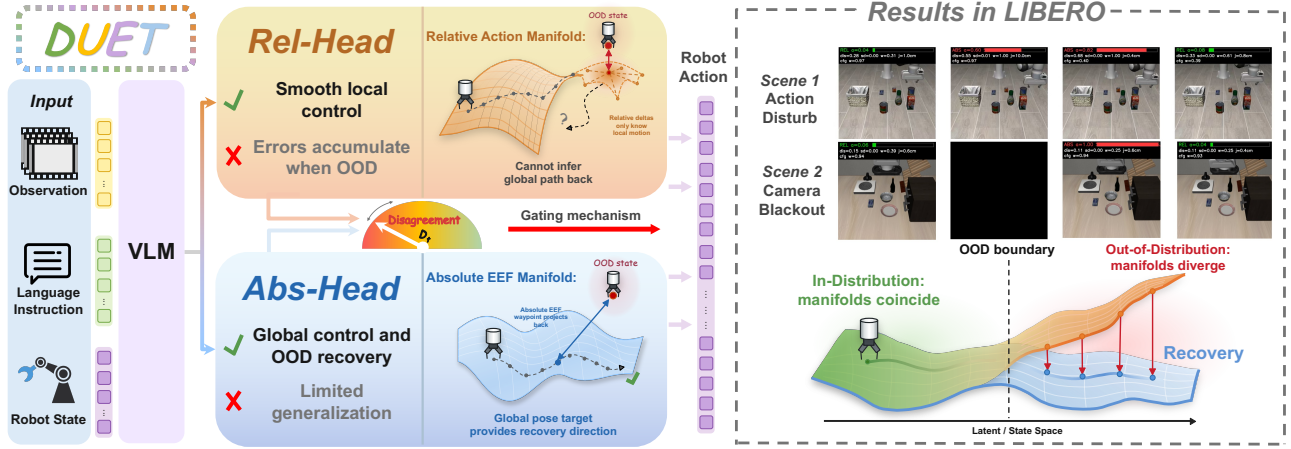


Fig. 1. DUET overview. Cross-space disagreement detects deviations and gates relative control with absolute-waypoint recovery.

II. RELATED WORK

A. Runtime Failure Detection

Runtime monitoring must distinguish unusual observations from deviations that threaten task completion. Sentinel combines temporal action consistency for erratic behavior with VLM-based assessment of task progress, addressing failures that remain locally consistent [8]. Without failure examples, FAIL-Detect frames monitoring as sequential OOD detection using scalar uncertainty signals and conformal calibration [3]. FIPER combines representation novelty with action-chunk entropy [9], while VLA-FAIL pairs feature-space distance with consistency across overlapping action chunks [10]. These approaches show why complementary signals can be more informative than novelty alone.

Supervised and semantic monitors offer a different trade-off. SAFE learns a multitask failure score from VLA features using successful and failed rollouts [11]; AHA is fine-tuned on procedurally generated failure trajectories to provide natural-language diagnoses [12]. DUET instead compares relative actions with transported absolute waypoints learned from successful demonstrations. Unlike a scalar alarm or diagnosis alone, this comparison also exposes a candidate correction. It targets recoverable execution drift, rather than general semantic failure recognition.

B. Recovery from Execution Failures

One route to recovery expands the training distribution: DAgger queries experts on policy-induced states [4], intervention-based learning uses human feedback and takeovers [13], [5], and IntervGen generates corrective trajectories from a small intervention set [6]. RACER augments demonstrations with recovery trajectories and rich language annotations, then uses a VLM supervisor to guide a visuomotor actor [14]. Optimized VLM prompts can also support motion correction and task-level recovery without this actor-training pipeline [15]. These methods address broader corrective behavior through additional supervision or semantic reasoning.

Recovery from existing demonstrations is also well established. Back to the Manifold uses an approximate training-data density and an equivariant visual-to-latent mapping to return from OOD states [16]. OCR derives an inverse recovery policy from an object-keypoint manifold without collecting additional recovery data [17]. Rewind-IL couples action-chunk inconsistency with respawning at VLM-verified demonstration checkpoints [18]; FAR uses failure-derived preferences and action perturbations for test-time adaptation and retry [19]. Thus, avoiding recovery demonstrations is not unique to DUET. Its distinction is to use jointly learned relative and absolute predictions for both the trigger and an online waypoint correction, without checkpoint retrieval or test-time parameter updates.

C. Action Representations for Robust Control

Action-space design changes both learning and deployment behavior. Aljalbout et al. compare control spaces for reinforcement learning and sim-to-real manipulation, showing that representation choice materially affects performance [20]. Feng et al. systematically study manipulation policies and report benefits of appropriate delta prediction, alongside complementary properties of joint- and task-space control [21]. These findings motivate treating representation as a design choice, but do not establish a recovery guarantee. DUET retains two operational-space views of the same demonstrations: relative commands for nominal control and absolute waypoints as return targets. Their disagreement is used at runtime, rather than selecting a single representation solely for nominal accuracy.

III. METHOD

A. Problem Formulation

At time t , the robot observes multi-view images o_t , a language instruction l , and proprioceptive state s_t . Trained only on successful demonstrations, DUET predicts the same

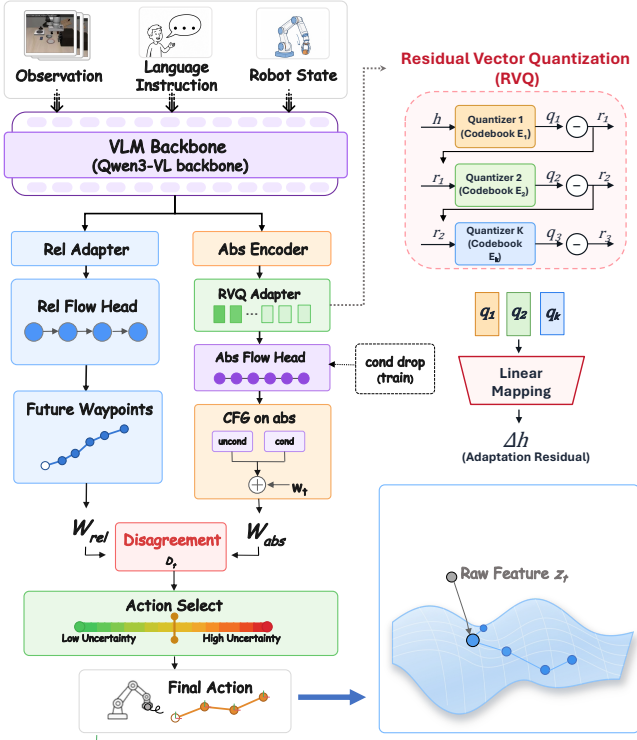


Fig. 2. DUET architecture: shared encoder, relative and RVQ-anchored absolute heads, and disagreement-gated recovery.

behavior in two action representations over a horizon K :

$$\hat{\Delta}A_t = f_{\text{rel}}(o_t, l, s_t) \in \mathbb{R}^{K \times 7}, \quad (1)$$

$$\hat{W}_t = f_{\text{abs}}(o_t, l) \in \mathbb{R}^{K \times 7}. \quad (2)$$

ΔA_t contains relative operational-space commands and W_t absolute waypoints $[x, y, z, r_x, r_y, r_z, g]$. Recorded deltas and future end-effector states supervise the respective heads. Fusion acts on the 6-D pose component a_p ; the deployed gripper command comes from the relative head.

Paired labels describe the same demonstrated continuation, letting both heads learn without annotating failures or recovery actions. At deployment, the live pose converts an absolute target into a correction, although it is not an explicit input to the absolute predictor.

B. Two Views of One Behavior

We use geometry as intuition, not as a formal guarantee. For instruction l , successful end-effector poses occupy a demonstrated support $\mathcal{M}_l \subset SE(3)$. The relative head answers *how to move next* from live pose p_t ,

$$\hat{a}_{p,t} = f_{\text{rel}}(o_t, l, s_t)_p, \quad (3)$$

which is locally precise but provides no return destination off support. The absolute head answers *where to be*,

$$\hat{w}_{p,t} = f_{\text{abs}}(o_t, l)_p, \quad (4)$$

providing a coarser target anchored by successful demonstrations.

We map the waypoint to a one-step correction at the live pose,

$$c_t = \Psi_{p_t}(\hat{w}_{p,t}), \quad (5)$$

instantiated in Eq. (9) by normalized translation and geodesic rotation residuals. This is a clipped first-order surrogate, not parallel transport on $SE(3)$.

Raw deltas and absolute coordinates cannot be compared directly: a robot displacement changes the correction required to reach a waypoint even when that waypoint remains unchanged. Expressing both predictions in local command units separates this execution mismatch from their different output parameterizations.

On support, c_t and $\hat{a}_{p,t}$ describe the same behavior and should broadly agree. Off support, local extrapolation can diverge from the demonstrated anchor. We use their residual as a heuristic—not calibrated—failure score and c_t as its implied recovery direction. Disagreement is neither necessary nor sufficient for failure: corrupted observations can degrade both heads together, as handled in Sec. III-E. We therefore validate it empirically in Sec. IV-D.

C. Dual-Head Architecture

Figure 2 shows the shared VLM and two flow-matching heads [22], [7]. The relative head attends to the full sequence and receives s_t . The absolute head excludes state tokens from pooling, preserving a complementary observation-conditioned reference rather than a second state-conditioned local controller.

It pools its state-free hidden sequence H^0 and applies residual vector quantization (RVQ) [23], [24]:

$$Z_q = \text{RVQ}(\text{Pool}(H^0)). \quad (6)$$

RVQ maps conditions to a discrete atlas of successful demonstrations. Its residual norm is auxiliary telemetry. Classifier-free guidance (CFG) [25] softens recovery predictions toward the demonstrated waypoint marginal.

Here, the atlas is a conditioning bottleneck, not a lookup of recorded recovery trajectories. Quantized features condition the absolute head; the head still predicts the waypoint. This distinction matters off support: quantization encourages reuse of learned structure but does not guarantee that the predicted target is reachable or task-correct.

D. Disagreement-Preserving Learning

Both heads train on paired labels from every successful trajectory, with flow-matching objectives and an RVQ commitment loss:

$$\mathcal{L} = \lambda_{\text{rel}}\mathcal{L}_{\text{rel}} + \lambda_{\text{abs}}\mathcal{L}_{\text{abs}} + \lambda_{\text{vq}}\mathcal{L}_{\text{vq}}, \quad (7)$$

where absolute labels are normalized future states and relative labels are recorded deltas. We omit cross-head consistency loss to preserve useful disagreement.

Agreement on successful behavior is encouraged indirectly by paired supervision. Explicitly forcing agreement on perturbed inputs could suppress the very residual used for monitoring; the separate objectives instead allow the

branches to respond differently when their conditioning becomes unreliable.

Asymmetric visual augmentation specializes the branches. The absolute head and RVQ adapter train on clean and perturbed observations so the global anchor remains stable under degraded input. The relative head is optimized only on the clean subset $\mathcal{B}_{\text{clean}}$:

$$\mathcal{L}_{\text{rel}} = \frac{1}{|\mathcal{B}_{\text{clean}}|} \sum_{i \in \mathcal{B}_{\text{clean}}} \ell_{\text{FM}}(\hat{\Delta} a^{(i)}, \Delta a^{(i)}). \quad (8)$$

This keeps the relative head sensitive and the absolute branch robust. For CFG, the absolute head uses learned null tokens on a subset of samples.

E. Joint Failure Detection and Recovery

At deployment the policy returns both unnormalized action chunks. With live pose $p_t = (x_t, R_t)$ and absolute waypoint $\hat{w}_{p,t} = (\hat{x}_t, \hat{R}_t)$, the transport map in Eq. (5) is

$$c_t = \text{clip}\left(\left[\begin{array}{c} (\hat{x}_t - x_t)/\sigma_p \\ \text{Log}(\hat{R}_t R_t^{-1})/\sigma_R \end{array}\right], -1, 1\right), \quad (9)$$

where $\text{Log} : SO(3) \rightarrow \mathfrak{so}(3)$ is the geodesic rotation correction and σ_p, σ_R convert pose error into action units. Cross-space disagreement is the primary failure signal,

$$D_t = \frac{\|c_t - \text{clip}(\hat{\delta} a_{p,t}, -1, 1)\|_2}{\sqrt{6}}. \quad (10)$$

The factor $\sqrt{6}$ makes D_t a root-mean-square command discrepancy across pose dimensions. Since both compared vectors are clipped componentwise, $D_t \in [0, 2]$; it is a discrepancy scale, not a failure probability. Translation and rotation are compared only after conversion by their respective action scales. A coarse state-distance term measures normalized violation of the training-state bounds:

$$S_t = \text{mean} \left[\frac{\max(s_{\min} - p_t, 0)}{s_{\max} - s_{\min}} + \frac{\max(p_t - s_{\max}, 0)}{s_{\max} - s_{\min}} \right]. \quad (11)$$

The failure score and recovery weight are $q_t = D_t + \lambda S_t$ and $\alpha_t = \sigma(k(q_t - \tau))$. DUET interpolates local execution and global recovery while keeping relative gripper control:

$$a_t = [a_{p,t}, a_{g,t}], \quad (12)$$

$$a_{p,t} = (1 - \alpha_t) \hat{\delta} a_{p,t} + \alpha_t c_t, \quad (13)$$

$$a_{g,t} = \hat{\delta} a_{g,t}, \quad (14)$$

Relative control dominates on support; rising q_t shifts the pose command toward c_t , and control returns after realignment. The previous fusion weight also sets the next absolute-head CFG scale, $w_t = w_{\max} - (w_{\max} - w_{\min})\alpha_{t-1}$. Thus D_t determines *when* and c_t *how* to recover. If vision is invalid, a guard freezes the last reliable waypoint and holds $\alpha_t=1$ until it returns.

TABLE I
FAILURE FAMILIES AND AFFECTED OOD QUANTITIES.

Family	Protocol	What becomes OOD
Action execution	Action perturbation (step 5)	Command-to-motion map
Robot displacement	EEF teleport (step 39)	Robot-arm pose
Object / grasp	Vertical grasp drop (step 67)	Robot-object configuration
Visual occlusion	Camera blackout (step 23)	Current visual observation
Contact / scene	External force	Robot-world interaction

IV. EXPERIMENTS

We organize the simulation experiments around three questions:

- (Q1) Failure-aware retry:** Can DUET convert execution failures into successful task completion?
- (Q2) Baseline comparison:** Does DUET improve robustness without sacrificing nominal performance?
- (Q3) Endogenous OOD detection:** Can cross-space disagreement detect failures using only successful demonstrations?

Closed-loop success is the primary outcome. We evaluate disturbed execution, nominal performance, and detection quality, then use ablations and hardware experiments to test the mechanism and its applicability across backbones.

A. Evaluation Protocol

We evaluate four LIBERO suites [26], each with 10 tasks and 50 initial states per task, using suite-specific 10k-step checkpoints trained only on successful demonstrations.

StarVLA [2] is the external relative-action baseline. DUET-Rel ($\alpha_t=0$) and DUET share a checkpoint, so their difference isolates adaptive recovery, whereas StarVLA→DUET-Rel captures dual-head co-training. Absolute-only ($\alpha_t=1$) and fixed fusion ($\alpha_t=0.5$) are inference-time ablations. DUET uses Qwen-VL features (size 2048), 16-/8-layer relative/absolute DiT-B heads, and 16 RVQ tokens from three codebooks (size 512, dimension 256), with $p_{\text{aug}}=0.3$, $p_{\text{drop}}=0.15$, $\lambda=1$, $\tau=0.5$, and $k=8$.

Perturbations use deterministic seeds and matched (task, initial state) keys. Fixed steps place each disturbance in a meaningful phase: initial reach (action, step 5), transport (teleport, 39), after gripper closure (drop, 67), and approach (blackout, 23). This ensures identical interventions across methods but samples only one recovery horizon per family; later injection leaves less time to recover, as reflected by Spatial teleport. We retain episodes ending before injection, report coverage for semantic preconditions, and use paired McNemar tests. Table I summarizes the families; external force is evaluated separately.

B. (Q1) Failure-Aware Retry

Table II reports rollout success, not success conditioned on detection. Against shared-checkpoint DUET-Rel, DUET improves action perturbation from 27.9% to 64.6%, displacement from 34.5% to 59.4%, grasp loss from 89.2% to 95.2%, and occlusion from 73.5% to 90.8%, exceeding StarVLA in every aggregate. Disabling fusion at inference isolates online correction from co-training.

TABLE II

RETRY SUCCESS (%; $N=500/\text{CELL}$). OVERALL: MEDIUM+SEVERE, ALL SUITES. REL→DUET: MCNEMAR $p<10^{-3}$ EXCEPT SPATIAL TELEPORT (0.21/0.067) AND MEDIUM OBJECT BLACKOUT.

	Medium				Severe			
Suite	StarVLA	Rel.	DUET	Δ	StarVLA	Rel.	DUET	Δ
<i>(i) Action perturbation (step 5)</i>								
Spatial	60.0	54.0	74.0	+20.0	14.0	12.8	43.2	+30.4
Object	41.6	48.8	84.4	+35.6	7.8	7.8	60.6	+52.8
Goal	52.2	47.2	74.2	+27.0	18.8	10.6	52.6	+42.0
LIBERO-10	36.2	36.6	78.6	+42.0	6.2	5.4	49.0	+43.6
Overall	29.6	27.9	64.6	+36.7	<i>med+sev combined</i>			
<i>(ii) EEf teleport (step 39)</i>								
Spatial	55.4	54.6	57.6	+3.0	46.8	48.2	52.6	+4.4
Object	43.4	36.2	76.0	+39.8	21.2	20.8	68.4	+47.6
Goal	41.6	33.4	52.4	+19.0	30.4	28.2	50.6	+22.4
LIBERO-10	40.4	35.8	60.8	+25.0	21.4	18.4	56.8	+38.4
Overall	37.6	34.5	59.4	+24.9	<i>med+sev combined</i>			
<i>(iii) Vertical grasp drop (Object only, medium, step 67)</i>								
Object	90.6	89.2	95.2	+6.0	—			
<i>(iv) Camera blackout (step 23)</i>								
Spatial	81.0	88.0	94.4	+6.4	42.4	63.2	95.2	+32.0
Object	91.8	98.6	96.8	−1.8	64.2	95.0	97.2	+2.2
Goal	65.4	86.0	86.4	+0.4	35.4	65.8	85.8	+20.0
LIBERO-10	60.0	65.6	84.2	+18.6	24.4	25.6	86.2	+60.6
Overall	58.1	73.5	90.8	+17.3	<i>med+sev combined</i>			

Action and robot displacement. DUET gains 36.7 pp under action perturbation and 24.9 pp under teleport. The scene remains observable, consistent with the absolute branch supplying a return target when relative commands become unreliable. Gains persist at both severities but vary by suite: severe Object teleport improves by 47.6 pp, whereas Spatial gains only 3.0/4.4 pp ($p=0.21/0.067$, not significant). Pre-injection termination and the remaining recovery horizon limit interpretation of the latter result.

Grasp and vision. A 5 cm drop yields a significant 6.0 pp gain from an 89.2% baseline ($p=2.4\times 10^{-5}$; coverage $\geq 98.8\%$). Restoring arm pose need not restore a changed grasp. During blackout, the validity guard freezes the last reliable waypoint; its 17.3 pp gain is not evidence for disagreement detection alone. Object-medium is the sole nonsignificant regression (−1.8 pp), so intervention is not uniformly beneficial.

Contact. In 4,000 external-force episodes, DUET reaches 95.95% versus 92.22%, without suite-level regression. Figure 4 contrasts this smaller gain with the larger arm-execution improvements.

a) Mechanism evidence. Figure 5 shows transient recovery: arm-state disturbances raise disagreement and correction weight; after realignment, relative control resumes. World-state responses are weaker, while complete visual dropout invokes the guard. These traces support the switching mechanism, but do not imply that every disagreement peak predicts failure.

On Object teleport (Fig. 6; $n=484\text{--}498$), DUET closes 4.69 cm within 32 steps and re-enters the clean-trajectory neighborhood in 97.9% of episodes (median 16 steps), versus 0.71 cm, 62.5%, and 58 steps for DUET-Rel. Absolute-only

TABLE III
NOMINAL LIBERO SUCCESS (%). STARVLA-GR00T: JOINT 30K TRAINING; DUET: SUITE-SPECIFIC 10K.

Method	Spatial	Object	Goal	LIBERO-10	Avg.
$\pi_{0.5}$ (reported)	98.8	98.2	98.0	92.4	96.9
StarVLA-GR00T (reported)	97.8	98.8	97.4	92.0	96.5
DUET-Rel ($\alpha_t=0$)	98.0	99.2	92.0	86.0	93.8
DUET	98.8	99.4	97.6	91.8	96.9

reaches 99.6% re-entry but zero task success in this analysis. Geometric proximity is insufficient without resuming task-appropriate local actions.

Together these results answer **Q1**: DUET converts recoverable deviations into successful retries, while grasp changes and invalid vision bound disagreement alone.

C. (Q2) Baseline Comparison

Table III shows that adaptive fusion raises nominal success from 93.8% to 96.9% at fixed capacity and training data, improving every suite. Larger gains on Goal (92.0% $\rightarrow$ 97.6%) and LIBERO-10 (86.0% $\rightarrow$ 91.8%) suggest correction of naturally accumulating errors, although success rates do not identify their cause. DUET matches the strongest reported average; differing backbones and protocols make the external rows contextual, not controlled training-efficiency comparisons.

D. (Q3) Endogenous OOD Detection

Table IV isolates the signal that triggers recovery, comparing D_t with success-only detectors under matched protocols [3], [8]. Unlike these standalone scores, D_t is coupled to the transported correction c_t .

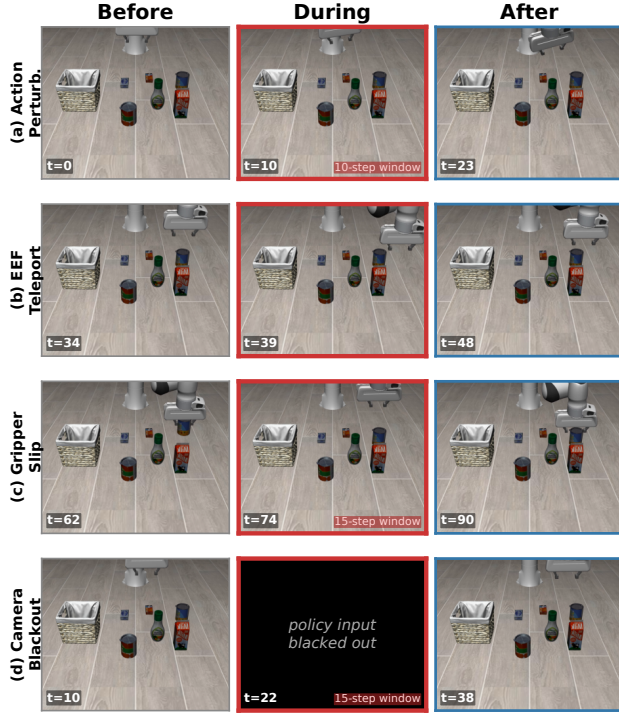


Fig. 3. Object-suite disturbances, before/during/after: (a) action perturbation, (b) EEf teleport, (c) grasp drop, (d) camera blackout.

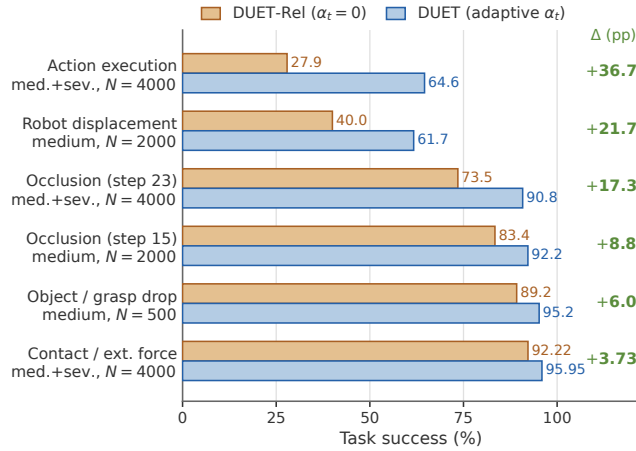


Fig. 4. Paired DUET-Rel→DUET gains by failure family.

DUET leads AUROC (0.845) and TPR (0.900), but RND leads TNR and logpZO localization. Better ranking therefore does not imply fewer false alarms or better timing at the evaluated threshold. All methods’ low TNR (0.24–0.36) indicates frequent nominal hard-threshold triggers.

DUET applies a continuous correction weight, not a binary stop command. Its clean mean $\alpha_t=0.139$ supports predominantly relative control, not the absence of false positives; costly unnecessary corrections would require a stricter operating point. Q1 evaluates the closed-loop benefit of coupling D_t to the paired correction c_t .

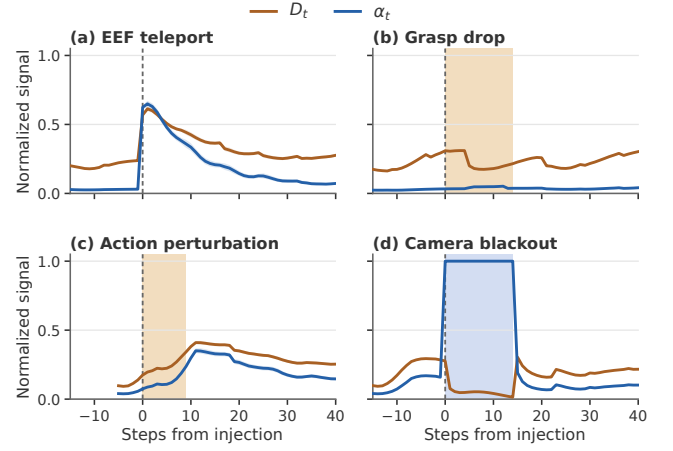


Fig. 5. Injection-aligned D_t, α_t on Object (mean, pointwise 95% CI): (a,c) arm-state OOD, (b) world-state change, (d) validity guard.

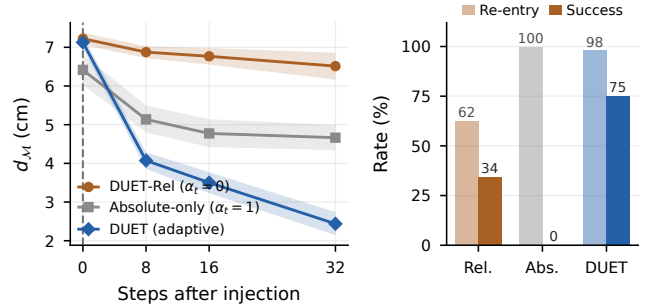


Fig. 6. Object teleport (medium, step 39; $n=484\text{--}498$). Left: clean-trajectory distance. Right: re-entry and task success.

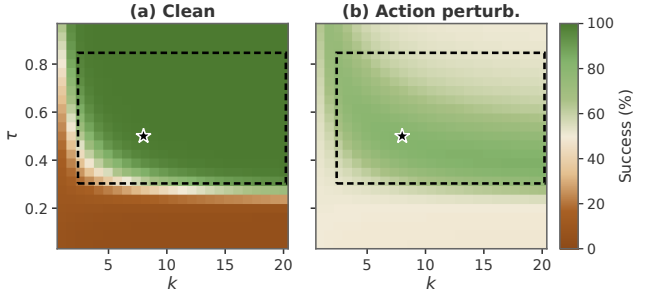


Fig. 7. Object τ, k sensitivity (500 traces/condition). Rectangle: clean $> 90\%$, OOD $> 75\%$; star: operating point.

E. Ablations

Table V tests whether adding the absolute action is sufficient. On Object, absolute-only and fixed fusion reach only 5.2% and 46.2% clean success. Fixed fusion also underperforms DUET-Rel under action perturbation (41.0% versus 48.8%), while adaptive fusion reaches 84.4%, supporting conditional waypoint use. The four-suite displacement average uses medium disturbances (40.0% $\rightarrow$ 61.7%); Table II’s 34.5% $\rightarrow$ 59.4% pools both severities.

Table VI separates CFG from training replacements. With-

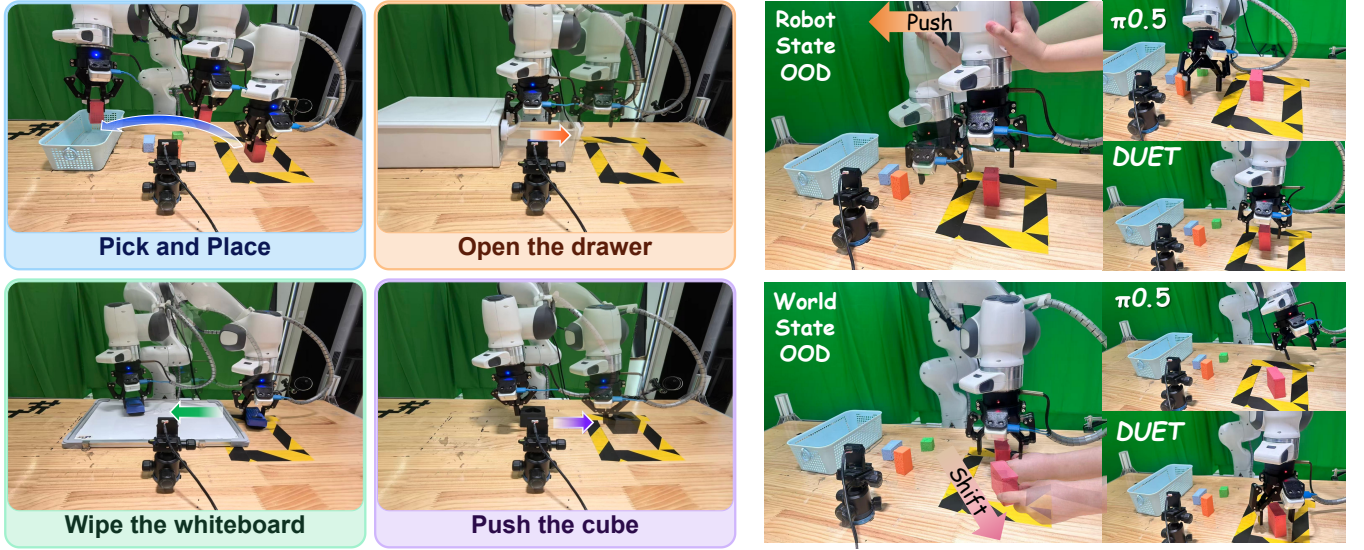


Fig. 8. Franka evaluation. Left: pick-and-place, drawer opening, wiping, and pushing. Right: pushing recovery under EEF (top) and object (bottom) shifts.

TABLE IV

MATCHED-PROTOCOL DETECTION. HIGHER IS BETTER; COLUMN BESTS IN BOLD.

Method	TPR $\uparrow$	TNR $\uparrow$	AUROC $\uparrow$	Loc.@16 $\uparrow$
logpZO [3]	0.889	0.243	0.817	0.569
RND [27], [3]	0.842	0.361	0.549	0.330
PCA-kmeans [3]	0.884	0.283	0.625	0.503
STAC [8]	0.839	0.311	0.494	0.260
DUET (D_t)	0.900	0.265	0.845	0.523

TABLE V

FUSION ABLATIONS (SUCCESS, %). FOUR-SUITE $N=2000/2000/4000$; MATCHED OBJECT $N=500$.

Controller	Gate	Clean	Robot displ.	Action exec.
<i>Four-suite average</i>				
DUET-Rel	$\alpha_t=0$	93.8	40.0	27.9
DUET	adaptive α_t	96.9	61.7	64.6
<i>Object suite, matched protocol</i>				
DUET-Rel	$\alpha_t=0$	99.2	36.2	48.8
Absolute-only	$\alpha_t=1$	5.2	1.1	0.8
Fixed fusion	$\alpha_t=0.5$	46.2	39.4	41.0
DUET	adaptive α_t	99.2	76.0	84.4

out adaptive CFG, Object displacement success falls from 76.0% to 52.6% and action-perturbation success from 84.4% to 57.2%, with lower clean success. On LIBERO-10, symmetric augmentation reduces all three outcomes, supporting branch-specific training. RVQ-to-MLP replacement yields zero success in this configuration, not a general rejection of continuous heads. Comparisons are within each suite-specific block.

a) Fusion on clean trajectories.: Across 2,000 clean episodes, mean α_t is 0.139, its 95th percentile is 0.339, and only 1.7% of steps exceed 0.5: relative control remains dominant.

TABLE VI

COMPONENT ABLATIONS (SUCCESS, %). CFG: OBJECT, $N=500$; RETRAINS: LIBERO-10, $N=500/500/1000$.

Variant	Clean	Robot displ.	Action exec.
<i>CFG schedule, Object suite</i>			
No adaptive CFG	88.2	52.6	57.2
Full DUET	99.2	76.0	84.4
<i>Training replacements, LIBERO-10</i>			
RVQ $\rightarrow$ MLP	0.0	0.0	0.0
Symmetric augmentation	86.8	58.6	60.9
Full DUET	91.8	60.8	63.8

b) Sensitivity to τ and k .: Figure 7 shows a broad Object region above 90% nominal and 75% OOD success containing $\tau=0.5, k=8$. Thus, the trade-off is not confined to one parameter pair, although this range is not validated across all suites or hardware tasks.

F. Real-Robot Evaluation

We test whether the DUET design applies to a separately trained π_0 (OpenPi) controller [7], rather than transferring simulation weights directly. A 7-DoF Franka Panda uses wrist and third-person cameras at 20Hz. Each task is fine-tuned for 5k steps from π_0 -base on 50 VR demonstrations. The four tasks are pick-and-place, pushing, drawer opening, and wiping (Fig. 8, left).

Each task is evaluated under nominal execution and two OOD families ($N=20$ trials per cell): a random-direction EEF shift (robot state) and an object displacement (world state), with task-specific intervention phases and magnitudes. Random directions are paired across π_0 and π_0 +DUET.

DUET preserves 90.0% aggregate nominal success (Table VII); pushing loses one trial and wiping gains one. Robot-state OOD improves from 10.0% to 77.5% across all four tasks, while world-state OOD rises from 6.3% to 58.8%. The remaining gap is consistent with changed object relationships

TABLE VII

FRANKA SUCCESS (%), π_0 -BASE; $N=20$ /CELL, 80 OVERALL). ROBOT:
RANDOM-DIRECTION EEf SHIFT; WORLD: OBJECT SHIFT. Δ : +DUET
MINUS π_0 .

Task	Nominal		Robot-state (EEf shift)			World-state (Obj. shift)		
	π_0	+DUET	π_0	+DUET	Δ	π_0	+DUET	Δ
Pick-and-place	95.0	95.0	15.0	85.0	+70.0	5.0	50.0	+45.0
Block pushing	95.0	90.0	10.0	75.0	+65.0	0.0	60.0	+60.0
Drawer opening	95.0	95.0	10.0	80.0	+70.0	20.0	70.0	+50.0
Whiteboard wiping	75.0	80.0	5.0	70.0	+65.0	0.0	55.0	+55.0
Overall	90.0	90.0	10.0	77.5	+67.5	6.3	58.8	+52.5

being harder to recover than arm alignment, but task-specific interventions preclude severity-matched comparison (Fig. 8). With 20 trials per cell, the results support applicability to these tasks and backbone, not broad cross-robot generalization. No additional detector or recovery dataset is used.

V. LIMITATIONS

DUET requires absolute-state labels aligned with relative chunks and may need hardware calibration. Its min-max state proxy can miss multimodal structure, while the stateless absolute branch can lose information not visible in images. D_t is weaker for world-state changes and uninformative when both heads share corrupted vision. The guard handles only the latter; irreversible failures and tasks requiring a new strategy remain out of scope.

VI. CONCLUSION

We presented DUET, a dual-branch VLA for endogenous failure detection and recovery. Comparing a local relative-action command with a stateless, RVQ-anchored absolute waypoint lets cross-space disagreement both flag OOD execution and supply a closed-loop recovery direction, without an external detector or additional recovery demonstrations. Because the module attaches only to the shared vision-language representation, the same design transfers from StarVLA in simulation to π_0 on a real Franka arm.

REFERENCES

- [1] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. Foster, T. Lampe, P. Sanketi, Q. Vuong, T. Kollar, B. Burchfiel, R. Tedrake, D. Sadigh, C. Finn, and P. Liang, “OpenVLA: An open-source vision-language-action model,” *arXiv preprint arXiv:2406.09246*, 2024.
- [2] StarVLA Community, “StarVLA: A lego-like codebase for vision-language-action model developing,” <https://github.com/starVLA/starVLA>, 2026.
- [3] C. Xu, T. K. Nguyen, E. Dixon, C. Rodriguez, P. Miller, R. Lee, P. Shah, R. Ambrus, H. Nishimura, and M. Itkina, “Can we detect failures without failure data? uncertainty-aware runtime failure detection for imitation learning policies,” in *Proceedings of Robotics: Science and Systems*, 2025.
- [4] S. Ross, G. Gordon, and D. Bagnell, “A reduction of imitation learning and structured prediction to no-regret online learning,” in *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, 2011.
- [5] H. Liu, S. Nasiriany, L. Zhang, Z. Bao, and Y. Zhu, “Robot learning on the job: Human-in-the-loop autonomy and learning during deployment,” in *Proceedings of Robotics: Science and Systems*, 2023.
- [6] R. Hoque, A. Mandlekar, C. R. Garrett, K. Goldberg, and D. Fox, “IntervenGen: Interventional data generation for robust and data-efficient robot imitation learning,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2024, pp. 2840–2846.

- [7] K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai, L. Groom, K. Hausman, B. Ichter, J. Jakubczak, T. Jones, A. Khazatsky, S. Levine, A. Li-Bell, Y. Lu, S. Nair, K. Pertsch, Q. V. Xu, F. Xia, and T. Xiao, “ π_0 : A vision-language-action flow model for general robot control,” *arXiv preprint arXiv:2410.24164*, 2024.
- [8] C. Agia, R. Sinha, J. Yang, Z. Cao, R. Antonova, M. Pavone, and J. Bohg, “Unpacking failure modes of generative policies: Runtime monitoring of consistency and progress,” in *Proceedings of the Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270, 2025, pp. 689–723.
- [9] R. Römer, A. Kobras, L. Worbis, and A. P. Schoellig, “Failure prediction at runtime for generative robot policies,” in *Advances in Neural Information Processing Systems*, 2025.
- [10] F. Seligmann, E. Gospodinov, E. U. Dincer, and G. Neumann, “VLA-FAIL: Efficient task failure detection for finetuned vision-language-action models,” *arXiv preprint arXiv:2606.21386*, 2026.
- [11] Q. Gu, Y. Ju, S. Sun, I. Gilitschenski, H. Nishimura, M. Itkina, and R. Shkurti, “SAFE: Multitask failure detection for vision-language-action models,” in *Advances in Neural Information Processing Systems*, vol. 38, 2025.
- [12] J. Duan, W. Pumacay, N. Kumar, Y. R. Wang, S. Tian, W. Yuan, R. Krishna, D. Fox, A. Mandlekar, and Y. Guo, “AHA: A vision-language-model for detecting and reasoning over failures in robotic manipulation,” in *International Conference on Learning Representations*, 2025.
- [13] J. Spencer, S. Choudhury, M. Barnes, M. Schmittle, M. Chiang, P. Ramadge, and S. Srinivasa, “Learning from interventions: Human-robot interaction as both explicit and implicit feedback,” in *Proceedings of Robotics: Science and Systems*, 2020.
- [14] Y. Dai, J. Lee, N. Fazeli, and J. Chai, “RACER: Rich language-guided failure recovery policies for imitation learning,” in *IEEE International Conference on Robotics and Automation*, 2025, pp. 15 657–15 664.
- [15] H. Chen, Y. Yao, R. Liu, C. Liu, and J. Ichnowski, “Automating robot failure recovery using vision-language models with optimized prompts,” *arXiv preprint arXiv:2409.03966*, 2024.
- [16] A. Reichlin, G. L. Marchetti, H. Yin, A. Ghadirzadeh, and D. Kragic, “Back to the manifold: Recovering from out-of-distribution states,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2022, pp. 8660–8666.
- [17] G. J. Gao, T. Li, and N. Figueroa, “Out-of-distribution recovery with object-centric keypoint inverse policy for visuomotor imitation learning,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2025.
- [18] G. Zheng, S. Seenivasan, M. Johnson-Roberson, and W. Zhi, “Rewind-IL: Online failure detection and state respawning for imitation learning,” *arXiv preprint arXiv:2604.16683*, 2026.
- [19] H. Hao, S. N. Syed, J. Ichnowski, and J. Schneider, “FAR: Failure-aware retry for test-time recovery and continual policy improvement,” *arXiv preprint arXiv:2607.01111*, 2026.
- [20] E. Aljalbout, F. Frank, M. Karl, and P. van der Smagt, “On the role of the action space in robot manipulation learning and sim-to-real transfer,” *IEEE Robotics and Automation Letters*, vol. 9, no. 6, pp. 5895–5902, 2024.
- [21] Y. Feng, J. Zheng, Z. Wang, D. Liu, J. Li, J. Pang, T. Wang, and X. Zhan, “Demystifying action space design for robotic manipulation policies,” *arXiv preprint arXiv:2602.23408*, 2026.
- [22] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, and S. Song, “Diffusion policy: Visuomotor policy learning via action diffusion,” *The International Journal of Robotics Research*, 2024.
- [23] A. van den Oord, O. Vinyals, and K. Kavukcuoglu, “Neural discrete representation learning,” in *Advances in Neural Information Processing Systems*, 2017.
- [24] N. Zeghidour, A. Luebs, A. Omran, J. Skoglund, and M. Tagliasacchi, “Soundstream: An end-to-end neural audio codec,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 2022.
- [25] J. Ho and T. Salimans, “Classifier-free diffusion guidance,” *arXiv preprint arXiv:2207.12598*, 2022.
- [26] B. Liu, Y. Zhu, C. Gao, Y. Feng, Q. Liu, Y. Zhu, and P. Stone, “LIBERO: Benchmarking knowledge transfer for lifelong robot learning,” in *Advances in Neural Information Processing Systems*, 2023.
- [27] Y. Burda, H. Edwards, A. Storkey, and O. Klimov, “Exploration by random network distillation,” in *International Conference on Learning Representations*, 2019.